

Cybersecurity

cron and Scheduled Jobs

Lesson 5.2



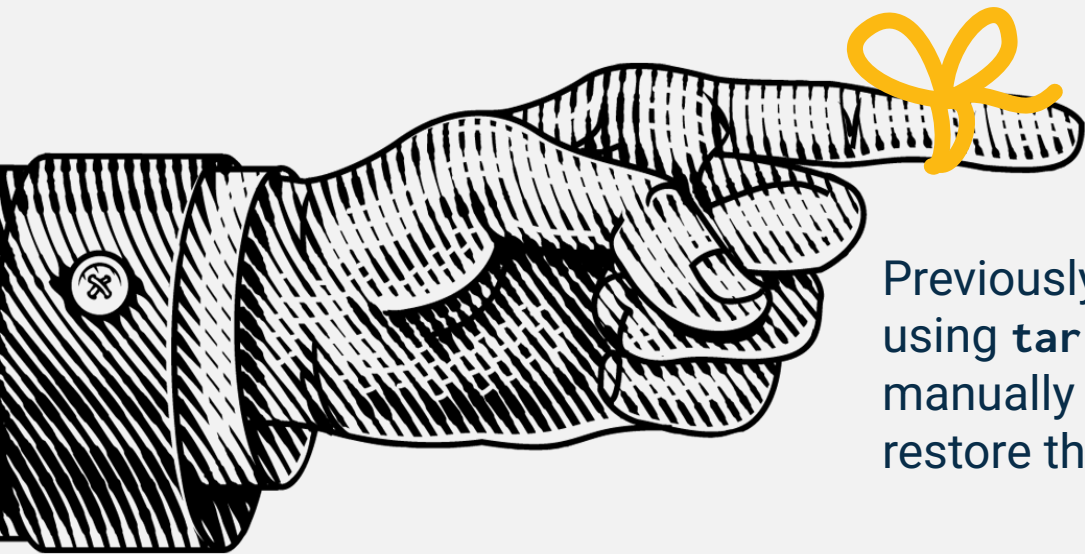
Learning Objectives

By the end of today's class, you will be able to:

- 1 Schedule regular jobs for individual users with crontab.
- 2 Write simple scripts for maintenance and security tasks.
- 3 Use **cron** to automate the execution of security scripts to perform maintenance.
- 4
- 5

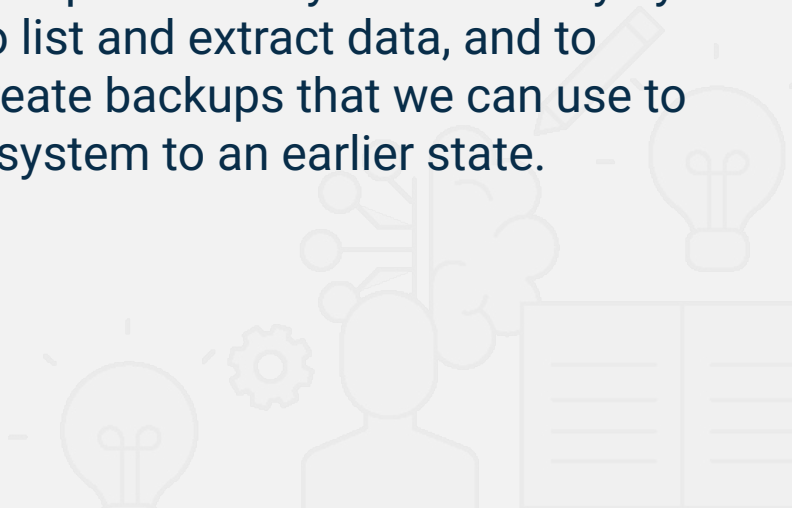


Welcome



Remember,

Previously, we protected system stability by using `tar` to list and extract data, and to manually create backups that we can use to restore the system to an earlier state.



Scheduling Backups

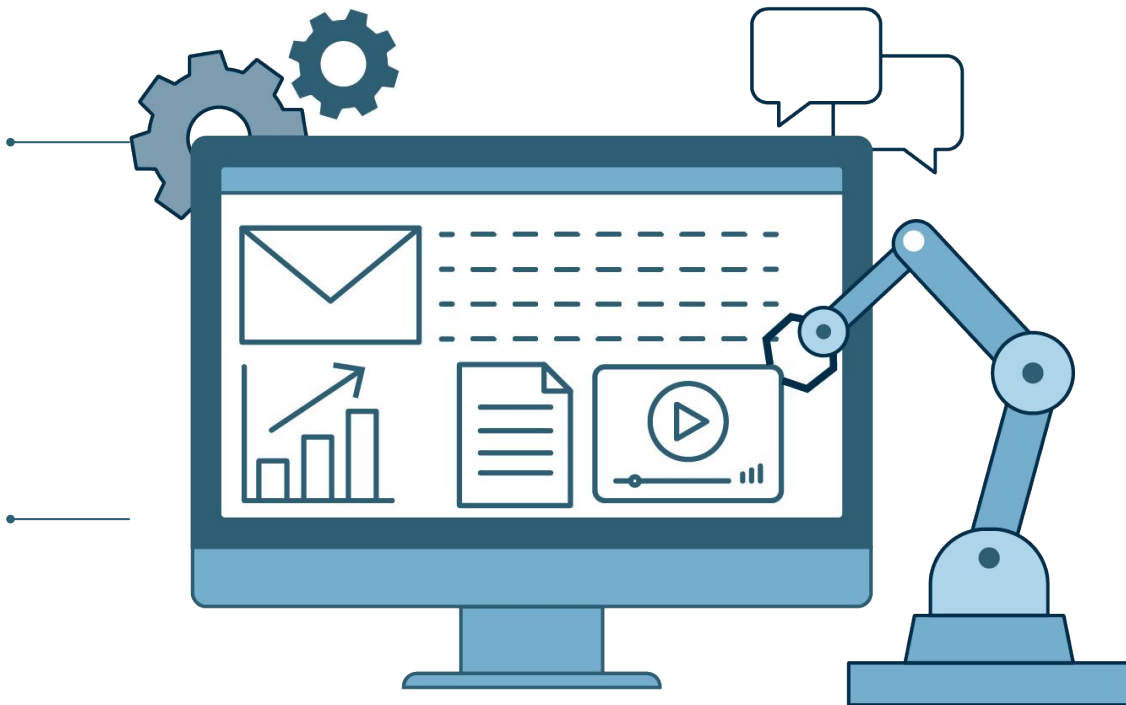
Using `cron` to schedule regular automation of tasks

Scripts

- Files that can contain multiple commands
- Executed by calling name

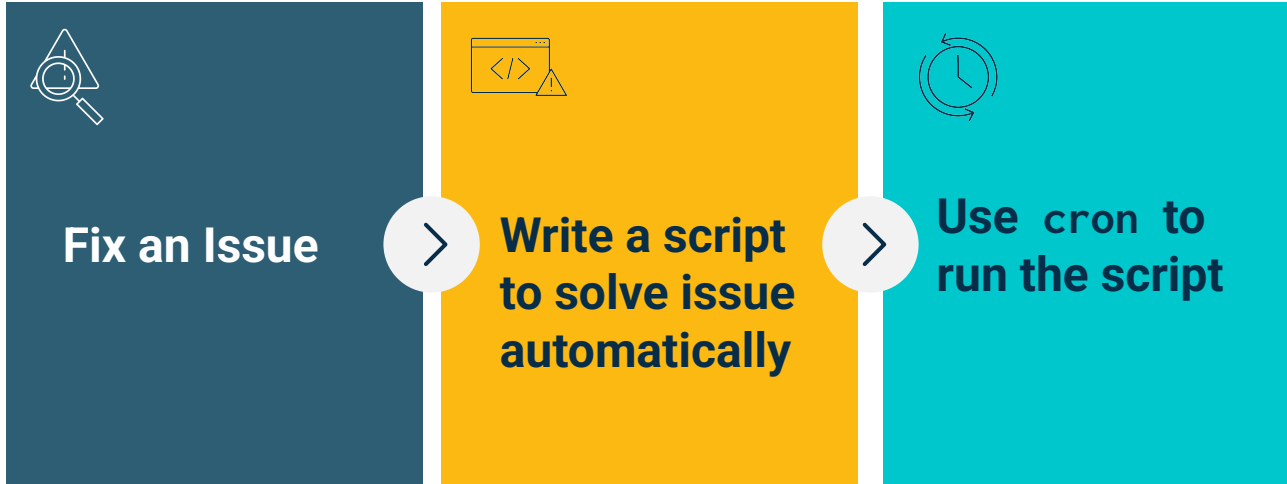
Scheduled Jobs

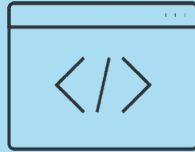
- Run commands or scripts at designated times



System Administration Workflow

Tools are often used together in a similar manner





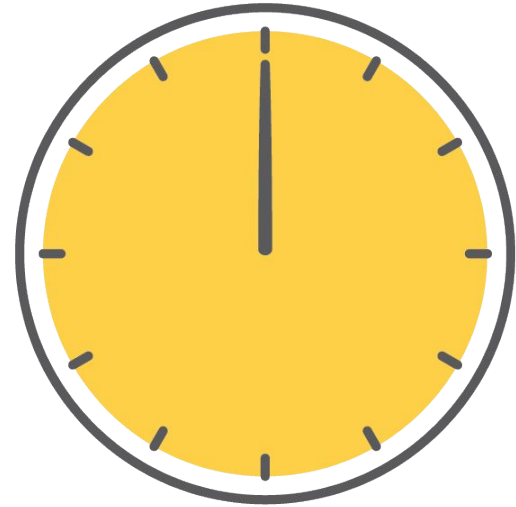
Overview of cron

Consider the Following Scenario

Before leaving work, you spend **10 minutes** deleting your cache and your trash bin, and backing up your documents folder.



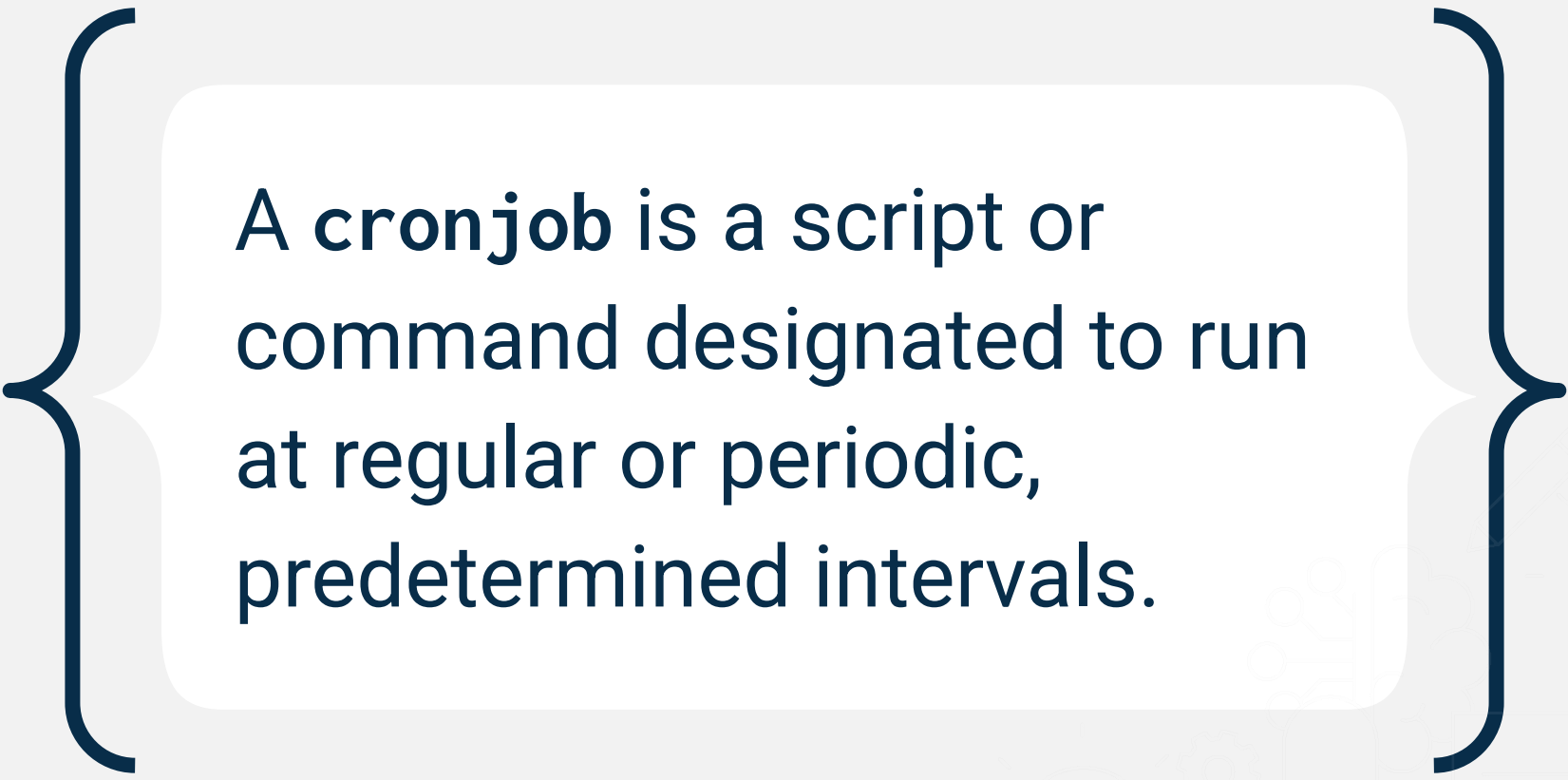
You also spend **one hour** per day installing software updates.



Introducing cron



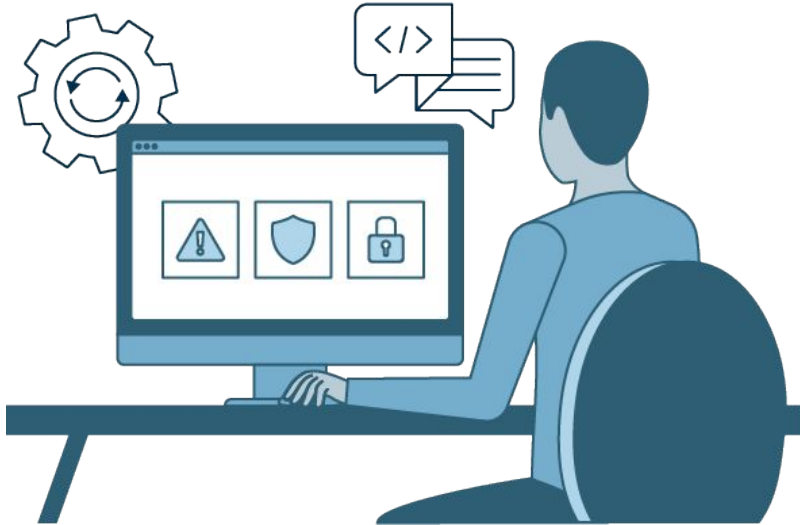
Rather than spending time manually completing these repeatable tasks, you can automate them using **cronjobs**.



A **cronjob** is a script or command designated to run at regular or periodic, predetermined intervals.

User-level

Automate the process of deleting cache, emptying trash, and backing up documents.



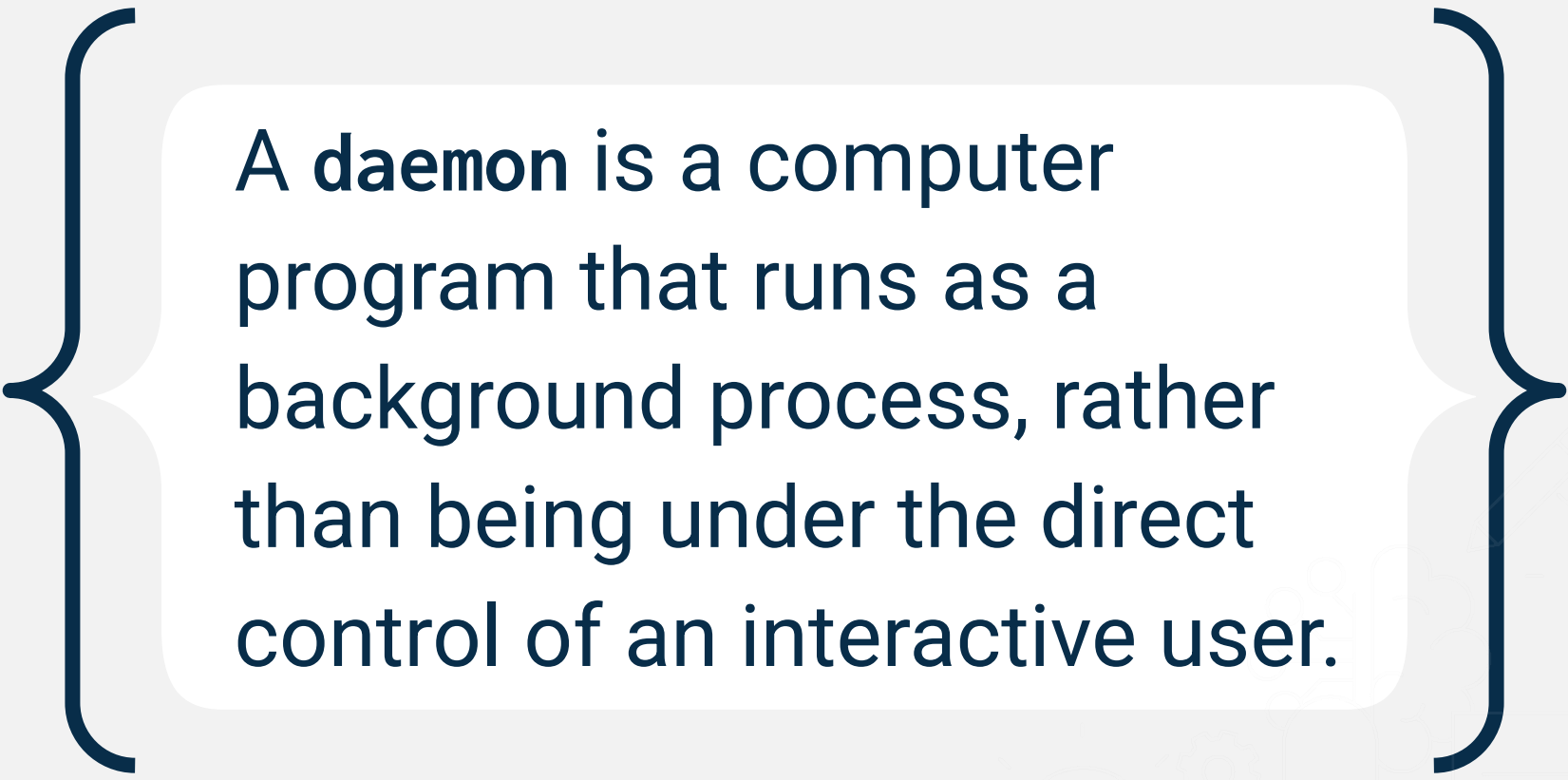
System-level

Automate daily software updates and installs.





The **cron** in **cronjob** refers to a system daemon that keeps track of when to run scheduled tasks.

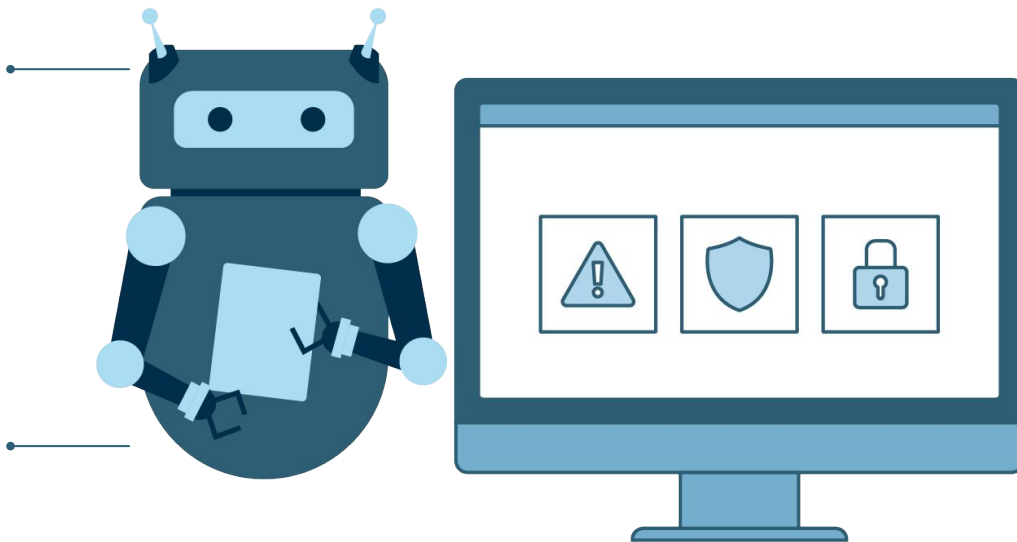


A daemon is a computer program that runs as a background process, rather than being under the direct control of an interactive user.

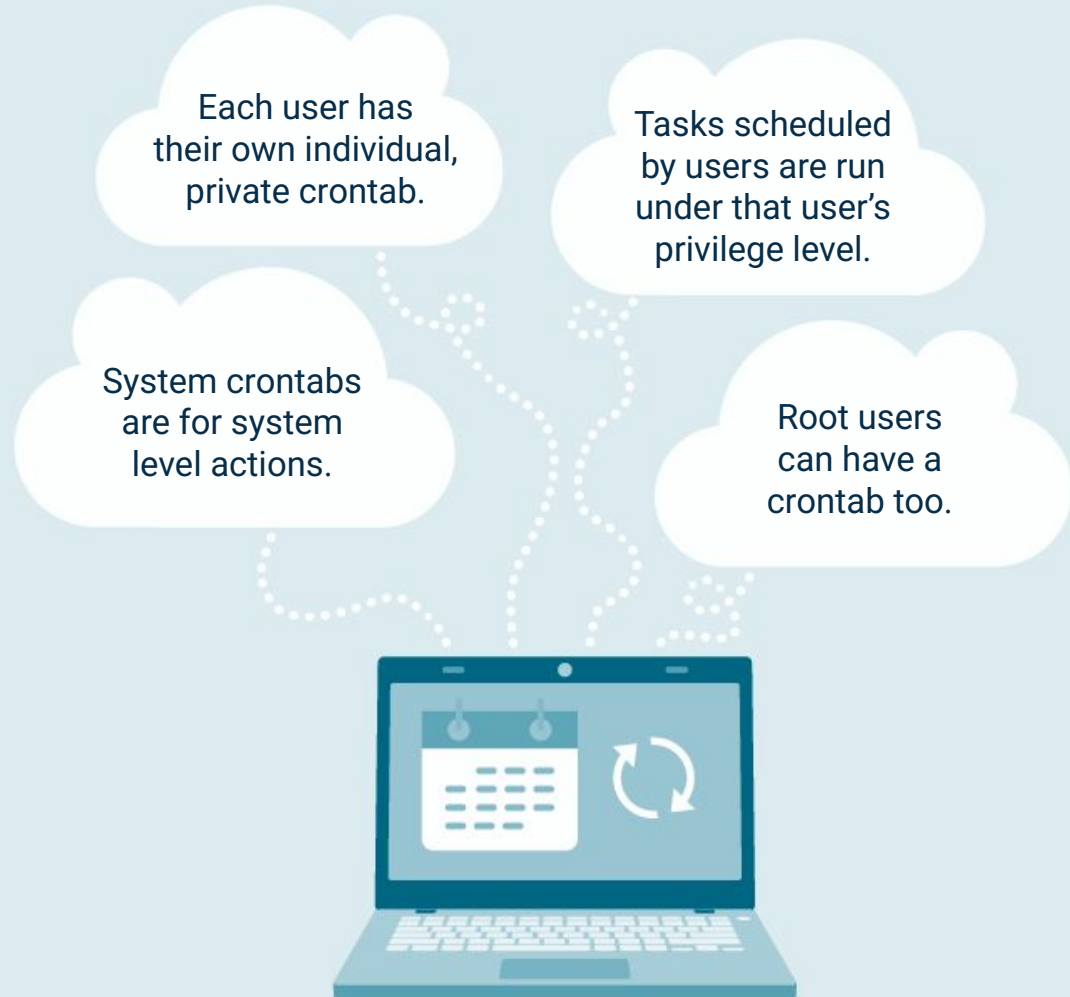
cron

cron is a robust task scheduler **daemon** that allows users to schedule repetitive tasks to run on a regular basis.

cron is most suitable for scheduling repetitive tasks.



Tasks are stored and scheduled in a file called a **crontab** (*cron table*).





cron Syntax

Viewing and modifying the crontab

List the contents of crontab

```
crontab -l
```

Edit the crontab

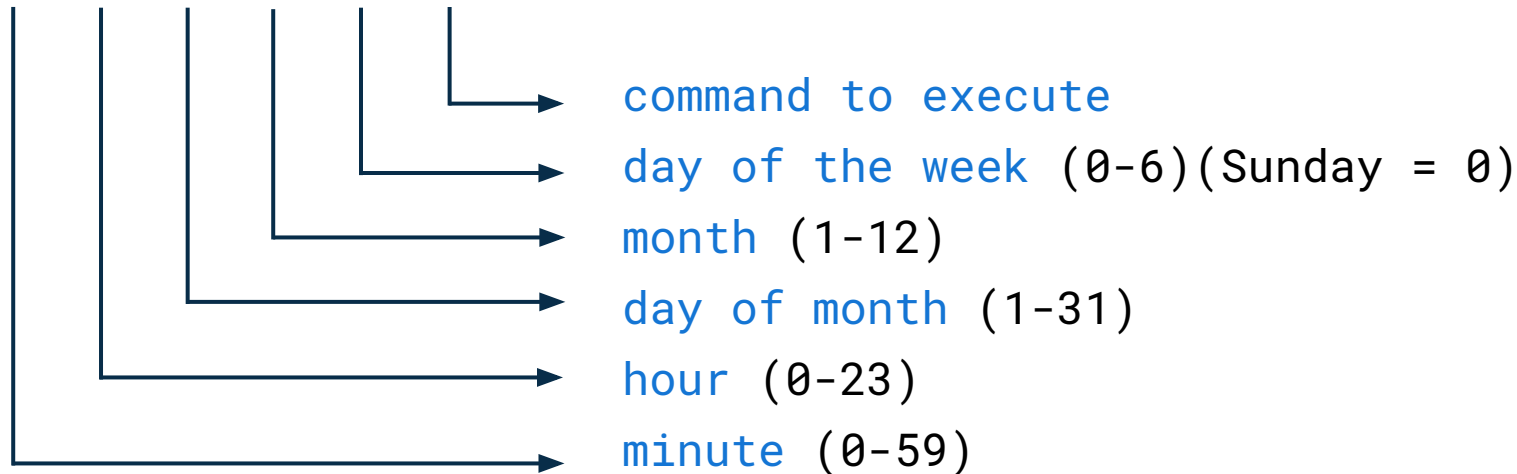
```
crontab -e
```

cron Example

Removes all files in the ~/Downloads directory at 11pm, every Saturday, every month

```
0 23 * * 6 rm ~/Downloads/*
```

```
0 23 * * 6 rm ~/Downloads/*
```





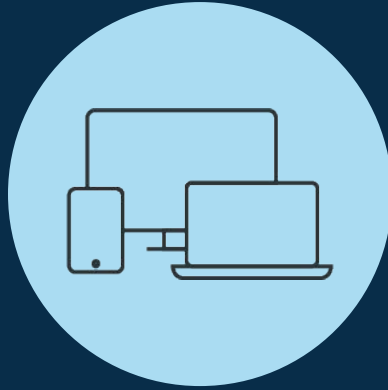
Demonstration Scenario

cronjobs run under the same permissions as the user who creates them. A **cronjob** created by user root will run with root privileges.

This introduces security risks. Anyone capable of privilege escalation can add a malicious cronjob to the root **crontab**.

- We'll revisit the risks of using **cron** with root privileges later in the unit.
- Best practice is to avoid using the root **crontab**.





Instructor Demonstration

crontab Generator



Activity:

Simple cronjobs

In this activity, you are a junior administrator at Rezifp Pharma Inc.

- There has been a wave of recent ransomware attacks. You will be responsible for using cron to automate backup jobs for the E-Prescription Treatment database.
- Rezifp Pharma Inc. maintains a large number of files associated with patients, doctors, and their treatments. Administrators at various clinics often create files that contain Personal Identifiable Information (PII) such as email addresses, passwords, biometric records, etc.

Suggested Time:
20 Minutes

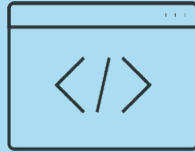


Time's up!
Let's review



Questions?





Introduction to **Scripts**

Why Scripts?

`cron` is useful for managing single function tasks

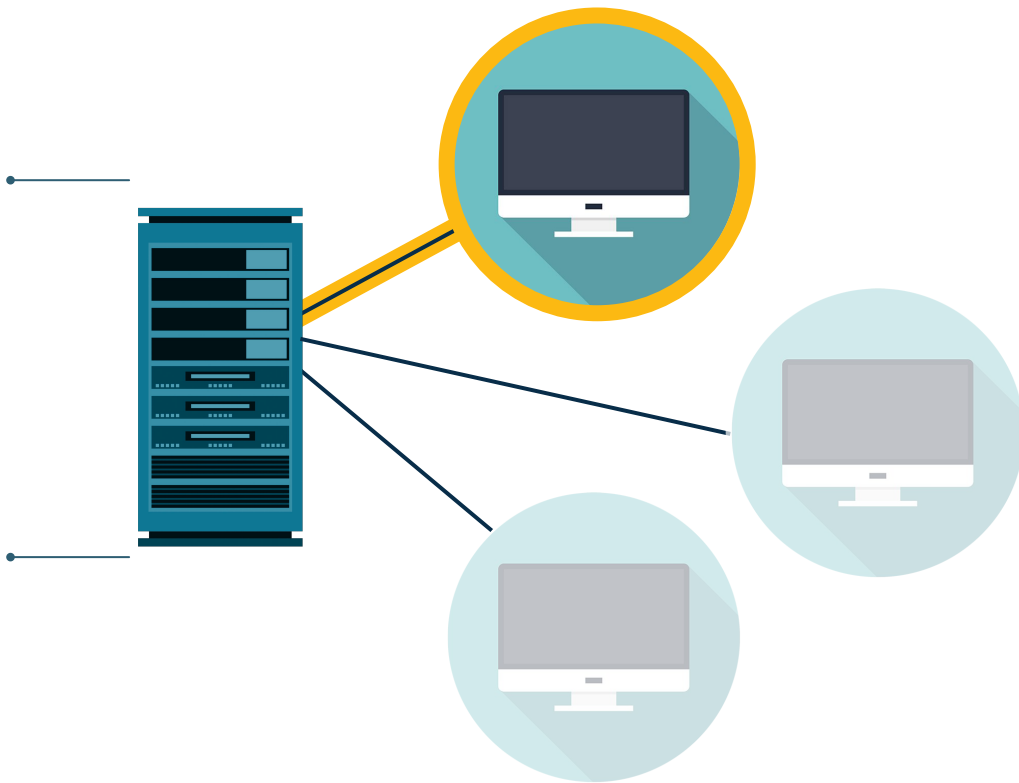
`cron` ideal for backing up a single directory:

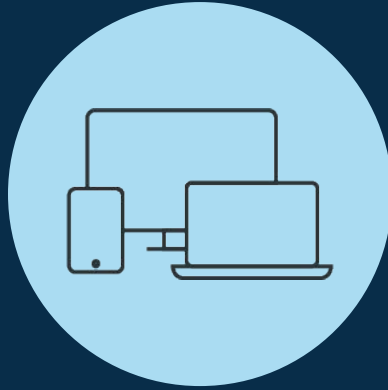
- ❖ `/home`

Not ideal for multiple directories:

- ❖ `/home/user1`

- ❖ `/home/user2`



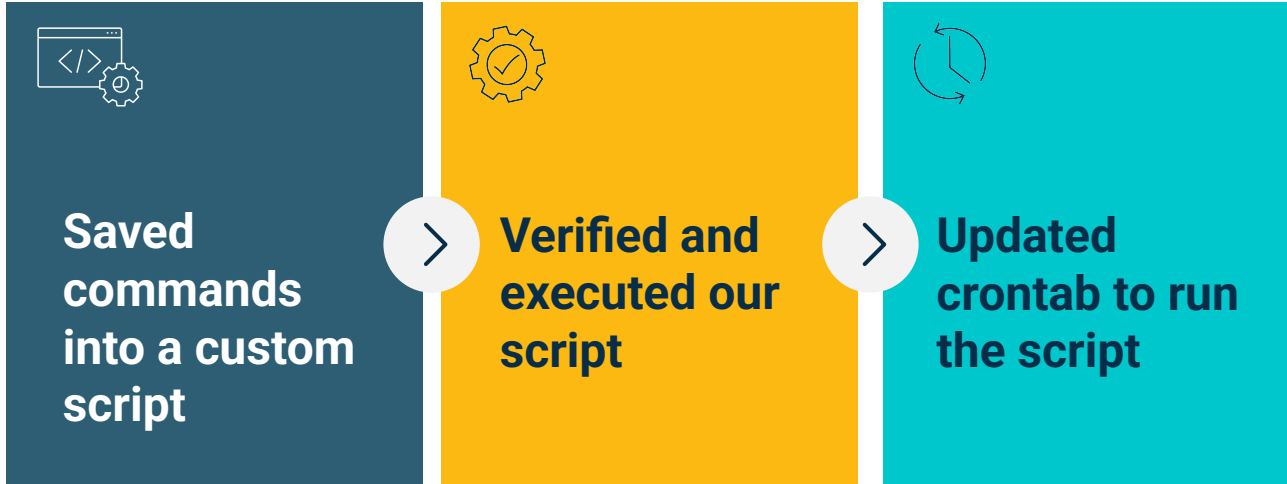


Instructor Demonstration

Scripting

Demo Summary

In the previous demo, we did the following:





Activity:

Introduction to Scripts

In this activity, you'll continue to play the role of a junior administrator at Rezifp Pharma Inc.

- In response to malware, adware, and virus attacks, your organization has asked you to ensure proper configuration management of all networking equipment and associated systems.
- Previously, you learned how to schedule backups using `cron`. Your senior systems administrator has requested that you begin automating these tasks by writing scripts.

Suggested Time:
30 Minutes



Time's up!
Let's review

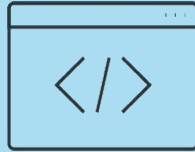


Questions?

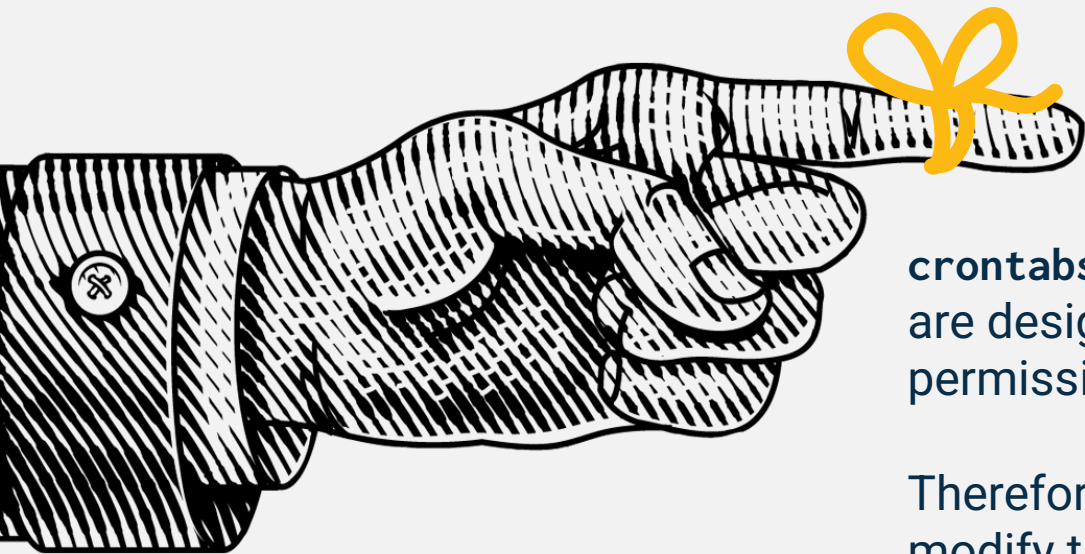




Break
15 mins



Scheduling Backups, Cleanups, and Security Checks



Remember,

`crontabs` belong to individual users and are designed to be run with the same permissions.

Therefore, they are typically unable to modify the system due to lack of `root` privileges.



System-Wide cron Directories

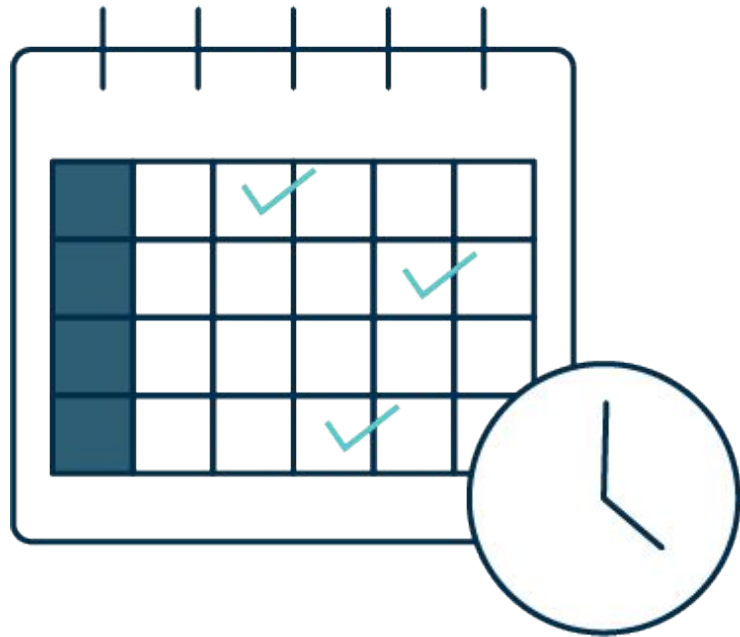
-System-wide cron directories:

- `/etc/cron.d`
- `/etc/cron.daily`
- `/etc/cron.weekly`
- `/etc/cron.monthly`

-cronjobs that run with root privileges

-Dedicated time intervals:

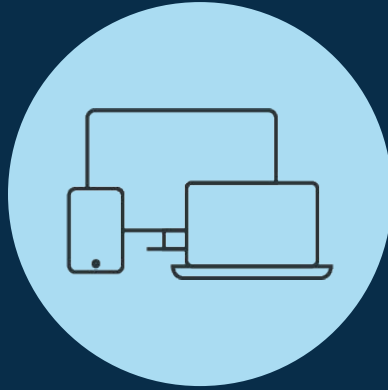
- `/etc/cron.weekly` runs once per week





Now we'll examine the
`/etc/cron` directories.





Instructor Demonstration

cron Directories



Lynis Scanner

Security scanner used to scan for vulnerabilities

Generates and saves reports

Numerous scan type options

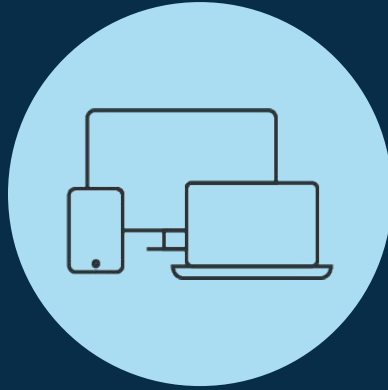
```
[+] Users, Groups and Authentication
-----
- Search administrator accounts...      [ OK ]
- Checking UIDs...                      [ OK ]
- Checking chkgrp tool...               [ FOUND ]
- Consistency check /etc/group file...   [ OK ]
- Test group files (grpck)...            [ OK ]
- Checking login shells...              [ WARNING ]
- Checking non unique group ID's...     [ OK ]
- Checking non unique group names...    [ OK ]
- Checking LDAP authentication support   [ NOT ENABLED ]
- Check /etc/sudoers file               [ NOT FOUND ]

[ Press [ENTER] to continue, or [CTRL]+C to stop ]

[+] Shells
-----
- Checking console TTYS...              [ WARNING ]
- Checking shells from /etc/shells...
  Result: found 6 shells (valid shells: 6).

[ Press [ENTER] to continue, or [CTRL]+C to stop ]

[+] File systems
-----
- [FreeBSD] Querying UFS mount points (fstab)... [ OK ]
- Query swap partitions (fstab)...       [ OK ]
- Testing swap partitions...             [ OK ]
- Checking for old files in /tmp...      [ WARNING ]
- Checking /tmp sticky bit...            [ OK ]
```



Instructor Demonstration

Lynis



Activity:

Scheduling Backups, Cleanups, and Security Checks

In this activity, you'll continue to play the role of a junior administrator at Rezifp Pharma Inc.

- Now that you've developed scripts to keep the system clean, up-to-date, and properly backed up, the senior security manager has requested that you begin using system-wide `cron` directories to schedule your scripts. Additionally, you have been asked to schedule regular security scans with Lynis.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Recap

cron and Scheduled Jobs

01

Scheduled regular jobs for individual users with `crontab`.

02

Write simple scripts for maintenance and security tasks.

03

Use `cron` to automate the execution of security scripts to perform maintenance on a regular basis.



Next class

bash Scripting



Thank you